

Joshua Yuen

Electrical Engineering and Physics Student | Space Systems Advocate |
Photonic Systems Enthusiast

Boston, MA
(302) 688 9416
yuen.jo@northeastern.edu

EDUCATION

Northeastern University, Boston, MA— *BSEE in Electrical Engineering and Physics* | August 2025 - April 2029 (Expected)

- Member of the John Martinson Honors Program
- GPA: 3.961/4.00 | Dean's List - All Semesters.

Delaware County Christian School (DCCS), Newtown Square, PA— *High School Diploma* | September 2021 - May 2025

- Graduated Highest Honor Roll. GPA: 5.26 / 5.0 scale. Co-Salutatorian.

Pennsylvania Governor's School for the Sciences, Carnegie Mellon University, PA | Summer 2024

- Selected as one of 72 students statewide for an intensive 5-week STEM program that studied topics including Discrete Mathematics, Special Relativity and Genome Editing.
- Co-authored research paper, "Hide and Seek: A Multimodal Approach to Human Detection" which investigated hardware enhancements to facial recognition technology under faculty mentorship.

Drexel University Materials Science Summer Institute, Philadelphia, PA—Lab Student | August 2024 - February 2025

- **Electrospun carbon nanofibers** to study how varying electric field strength during the creation process affected fiber diameter & analyzed fibers using scanning electron microscopy to determine the relationship between variables.

Skills

- **Circuits:** KiCad, Soldering & Circuit Assembly, Raspberry Pi. **CAD:** Fusion360, AutoCAD, Solidworks, Blender. **Programming:** LaTeX, Python.

Honors

- National Merit Scholar (Sponsored by Northeastern University).
- Honors Self-Directed Learner Badge, Honors Impact Badge, Honors Community Engagement Badge.
- Gerald F. Tonks Endowed Scholarship.
- Qualified for the VEX Robotics World Championship (2025); competed as one of a few top-tier teams representing the state of Pennsylvania at the international level in Dallas, TX.

PROFESSIONAL EXPERIENCE

Microscale Acoustic and Photonic Systems (MAPS) Lab, Boston, MA— *Undergrad Researcher* | November 2025 - Present

- Developing a modified **shot-noise limited Michelson Interferometer** to allow for better characterization of micro-electrical mechanical systems.

NEU Satellite Labs (NEUSL), Boston, MA— *NEU Satellite Lab*, Electrical Team Deputy | December 2025 - May 2026; *Solar Array Project*, Electrical Lead | September 2025 - May 2025

- Working as second-in-command under NEU Satellite Labs Electrical Team Lead to: coordinate and assign electrical projects, serve as a resource for club members seeking assistance with said projects, and instruct new members through the creation of a set onboarding curriculum (currently in development).
- Overseeing all electrical engineering work for NEUSL's **Solar Arrays project**; working to develop an open-source deployable solar array design for use on future missions.

Simply Code (Nonprofit), Philadelphia, PA— *Chief Technology Officer* | August 2024 - February 2025

- Developed and maintained infrastructure to support the nonprofit's mission and operations, including building and launching the organization's first website using WordPress.
- Coordinated with volunteers and led the Technology Department, overseeing the planning, execution, and delivery of key technology projects, including partnerships with other organizations in the region.

PROJECTS & PUBLICATIONS

Project: IMU-Based Hand Gesture Interface for 3D Design and Modeling | August 2025 - Present

- Developing a wearable glove-based input system using 9-axis IMU sensors to enable real-time hand gesture control and object manipulation in Blender 3D creation software
- Designing the software-hardware pipeline for motion tracking, gesture interpretation, and CAD integration.
- Exploring shape memory alloy-based haptic feedback to provide tactile responses to virtual object interactions.

Project: Deployable Solar Array for Small Satellites | September 2025 - May 2026

- Researching and developing open-source deployable solar arrays for CubeSat power generation.
- Worked to investigate power budgets and determine optimal configuration of cells on each panel.
- Created block diagram for power distribution and researched possible circuit schematics for integrating sensing arrays into the panel circuitry.

Project: Automation of Dechlorination Tools for Aqua Pennsylvania | August 2024 - May 2025

- Led development of automation solutions to improve water treatment efficiency that applied engineering principles to real-world challenges in partnership with **Aqua Pennsylvania**, a utility company.
- Co-created and taught an accredited class that developed and implemented real-world projects.

Publication: Hide and Seek: A Multimodal Approach to Human Detection | July 2024

- Investigated the results of integrating IR sensing, motion profiling, and mmWave sensing into an open source facial recognition software with the goal of increasing accuracy in detecting occluded faces and darker skin tones. Implemented a machine learning model to interpret results.
- Published in the 39th Volume of the Journal of the Pennsylvania Governor's School for the Sciences.